# Block practice · Respiration

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Write or complete the respiration equations

Explain how respiration releases energy and ATP transfers it

Set up and explain the indicator practical comparing carbon dioxide in breath

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# Practice workbook · Write or complete the respiration equations

Word and balanced symbol equations for aerobic respiration and the word equation for anaerobic respiration in humans.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-RESPIRATION-EQUATION

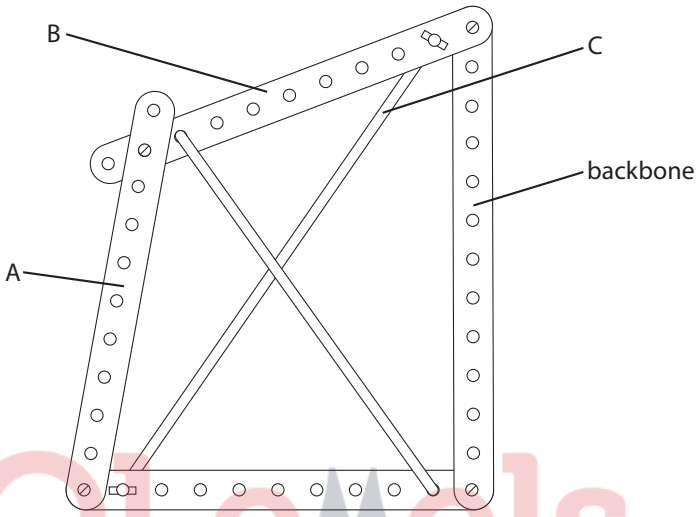
# 1. Question 5 · part b · 3 marks

June 2025 Paper 1

Worked-example retry · Attempt part b only.

**5** (a) The diagram shows a model that can be used to demonstrate the process of breathing.

The part that represents the backbone has been labelled.




(i) State which parts of the breathing system are represented by A, B and C.

**(3)**

A .....

B .....

C .....

(ii) State three other structures, not shown in the diagram, that are needed for breathing in the body.

**(3)**

1 .....

2 .....

3 .....

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# 1. Question 5 · part b · 3 marks

June 2025 Paper 1

Worked-example retry · Attempt part b only.

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(iii) Describe how the structures present in the thorax are used to allow a person to inhale.

**(6)**

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(b) Give the balanced chemical symbol equation for the process of aerobic respiration.

**(3)**

..... + ..... → ..... + ..... + .....

**(Total for Question 5 = 15 marks)**

## 1. Question 5 · part b · 3 marks · Working and answer

Working and answer

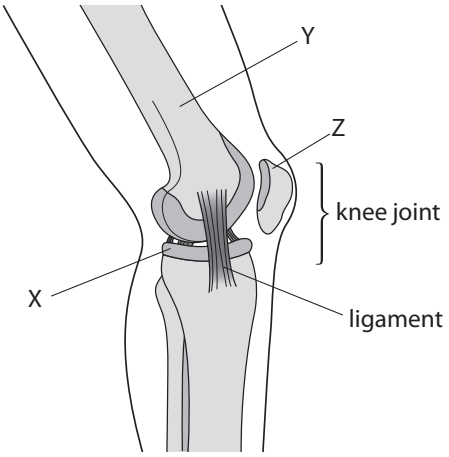
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## 2. Question 1 · part b · 2 marks

June 2019 Paper 1

Further practice · Attempt part b only.

**1** (a) The diagram shows structures in part of a human leg.




(i) The box lists structures in the human leg.

| | | | |
|---|---|---|---|
| cartilage | femur | kneecap | muscle |
| fibula | tendon | tibia | |

Use words from the box to name structures X, Y, and Z.

**(3)**

X .....

Y .....

Z .....

(ii) What is the function of ligaments in the knee joint?

**(1)**

- ☒ **A** ligaments attach muscles to the leg bones
- ☒ **B** ligaments hold the leg bones together
- ☒ **C** ligaments move the leg bones
- ☒ **D** ligaments stop the leg bones rubbing together

(iii) The type of joint found in the knee and the elbow is the same.

Give the name of this type of joint.

**(1)**

.....

**2**

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## 2. Question 1 · part b · 2 marks

June 2019 Paper 1

Further practice · Attempt part b only.

(b) Aerobic respiration releases energy for muscles to contract.

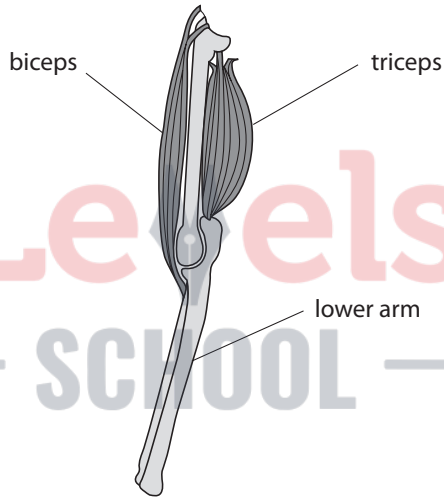
Complete the word equation for aerobic respiration.

**(2)**

glucose + ..... → ..... + water

(c) Diagram 1 shows a model of the human arm, with the lower arm extended.

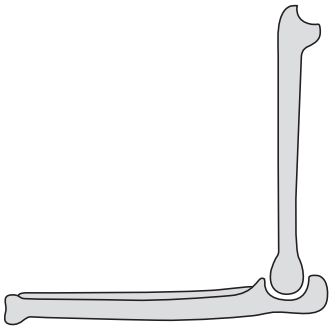
**Diagram 1**




Complete diagram 2 to show how the muscles change when the lower arm is raised.

**(2)**

**Diagram 2**



**(Total for Question 1 = 9 marks)**

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## 2. Question 1 · part b · 2 marks · Working and answer

Working and answer

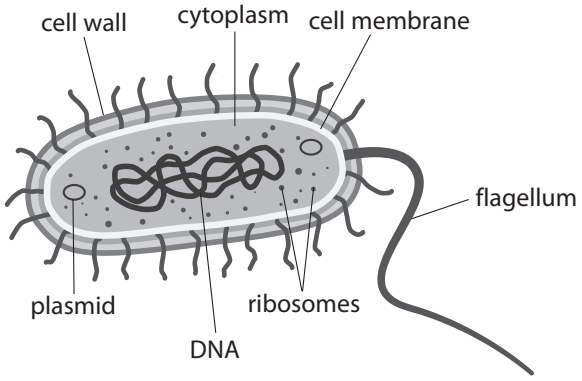
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# 3. Question 3 · part b.ii · 2 marks

January 2020 Paper 1

Further practice · Attempt part b.ii only.

**3** (a) The diagram shows a type of bacterial cell.




(i) The boxes give some structures found in the bacterial cell and some functions of cell structures.

Draw one straight line from each structure to its correct function.

**(3)**

| Structure | Function |
|---|---|
| cell membrane | protects against pathogens |
| | helps the cell to move |
| ribosomes | makes proteins |
| | holds the genetic code |
| flagellum | controls which substances enter and exit the cell |

(ii) State two structural differences between the bacterial cell and a sperm cell.

**(2)**

1 .....

.....

2 .....

.....

**8**

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# 3. Question 3 · part b.ii · 2 marks

January 2020 Paper 1

Further practice · Attempt part b.ii only.

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(b) Many bacteria absorb oxygen and nutrients from their environment. Oxygen and nutrients are used by bacteria in aerobic respiration.

(i) Which process transports oxygen into bacterial cells?

(1)

- ☒ **A** active transport
- ☒ **B** diffusion
- ☒ **C** excretion
- ☒ **D** osmosis

(ii) Complete the word equation for aerobic respiration.

(2)

oxygen + ..... → ..... + water

(iii) Explain why respiration is important for bacteria.

(2)

.....

.....

.....

.....

**(Total for Question 3 = 10 marks)**

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## 3. Question 3 · part b.ii · 2 marks · Working and answer

Working and answer

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# 4. Question 2 · part a · 4 marks

June 2021 Paper 2

Further practice · Attempt part a only.

**2** Aerobic respiration is the process that releases energy for use in the body.

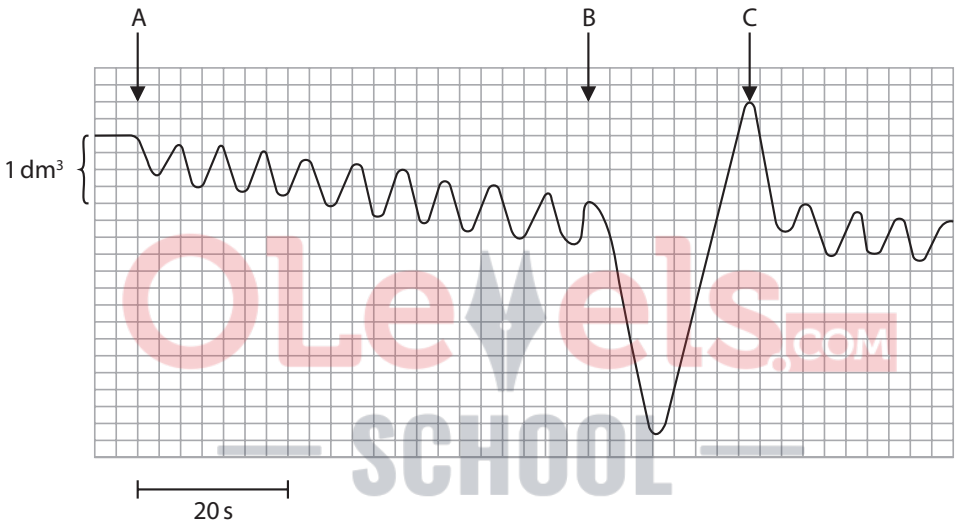
(a) Give the word equation for the process of aerobic respiration.

**(4)**

..... + ..... → ..... + ..... + energy

(b) A spirometer is used to measure the volumes of air breathed in and the volumes of air breathed out.

The spirometer trace shows the readings taken of a person at rest.




(i) Calculate the rate of breathing between points A and B shown on the trace.

**(2)**

rate = ..... breaths per minute

(ii) Calculate the difference in the volume of air breathed in and the volume of air breathed out between points B and C.

**(3)**

difference in volume = ..... $dm^3$

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P 6 6 4 3 3 A 0 4 2 4

## 4. Question 2 · part a · 4 marks · Working and answer

Working and answer

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# 5. Question 5 · part a.i · 2 marks

January 2023 Paper 2

Further practice · Attempt part a.i only.

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**5** Some athletes train at high altitudes to improve their performance.

Performance improves if more energy is released from aerobic respiration.

(a) (i) Complete the balanced symbol equation for aerobic respiration.

**(2)**

..... + $6O_2$ → ..... + $6CO_2$

(ii) Aerobic respiration produces ATP.

Describe how energy is released from ATP during exercise.

**(2)**

.....

.....

.....

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## 5. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

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## 6. Question 4 · part a.i · 2 marks

June 2023 Paper 2

Further practice · Attempt part a.i only.

**4** (a) (i) Complete the word equation for one type of respiration.

**(2)**

oxygen + ..... → ..... + .....

(ii) State the type of respiration shown by the equation.

**(1)**

.....

(iii) In which part of the cell does this type of respiration occur?

**(1)**

- ☒ **A** endoplasmic reticulum
- ☒ **B** mitochondrion
- ☒ **C** nucleus
- ☒ **D** ribosome

(iv) Name the molecule used to transfer energy in a cell.

**(1)**

.....

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## 6. Question 4 · part a.i · 2 marks · Working and answer

Working and answer

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## 7. Question 4 · part a.ii · 1 mark

June 2023 Paper 2

Further practice · Attempt part a.ii only.

**4** (a) (i) Complete the word equation for one type of respiration.

**(2)**

oxygen + ..... → ..... + .....

(ii) State the type of respiration shown by the equation.

**(1)**

.....

(iii) In which part of the cell does this type of respiration occur?

**(1)**

- ☒ **A** endoplasmic reticulum
- ☒ **B** mitochondrion
- ☒ **C** nucleus
- ☒ **D** ribosome

(iv) Name the molecule used to transfer energy in a cell.

**(1)**

.....

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## 7. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

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## 8. Question 7 · part a.i · 1 mark

June 2024 Paper 2R

Further practice · Attempt part a.i only.

**7** (a) Aerobic and anaerobic respiration both take place during exercise.

(i) Give the word equation for anaerobic respiration.

**(1)**

.....

(ii) Complete and balance the chemical symbol equation for aerobic respiration.

**(2)**

$C_6H_{12}O_6$ + ..... ⟶ ..... + ..... $CO_2$

(iii) A student investigates the rate of carbon dioxide production during exercise and at rest.

This is the equipment they use.

- four pieces of glass tubing
- two boiling tubes
- two rubber bungs
- mouthpiece
- limewater

Draw a diagram to show how the student should set up the equipment to carry out the investigation.

**(2)**

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**20**

P 7 4 4 9 5 A 0 2 0 2 8

## 8. Question 7 · part a.i · 1 mark · Working and answer

Working and answer

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## 9. Question 7 · part a.ii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part a.ii only.

**7** (a) Aerobic and anaerobic respiration both take place during exercise.

(i) Give the word equation for anaerobic respiration.

**(1)**

.....

(ii) Complete and balance the chemical symbol equation for aerobic respiration.

**(2)**

$C_6H_{12}O_6$ + ..... ⟶ ..... + ..... $CO_2$

(iii) A student investigates the rate of carbon dioxide production during exercise and at rest.

This is the equipment they use.

- four pieces of glass tubing
- two boiling tubes
- two rubber bungs
- mouthpiece
- limewater

Draw a diagram to show how the student should set up the equipment to carry out the investigation.

**(2)**

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## 9. Question 7 · part a.ii · 2 marks · Working and answer

Working and answer

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# Practice workbook · Explain how respiration releases energy and ATP transfers it

Why cells respire, measuring respiration rate, and the ADP/ATP cycle that passes energy to activities like contraction.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-RESPIRATION-ENERGY-ATP

# 1. Question 7 · part b · 5 marks

June 2024 Paper 2R

Worked-example retry · Attempt part b only.

**7** (a) Aerobic and anaerobic respiration both take place during exercise.

(i) Give the word equation for anaerobic respiration.

**(1)**

.....

(ii) Complete and balance the chemical symbol equation for aerobic respiration.

**(2)**

$C_6H_{12}O_6$ + ..... ⟶ ..... + ..... $CO_2$

(iii) A student investigates the rate of carbon dioxide production during exercise and at rest.

This is the equipment they use.

- four pieces of glass tubing
- two boiling tubes
- two rubber bungs
- mouthpiece
- limewater

Draw a diagram to show how the student should set up the equipment to carry out the investigation.

**(2)**

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**20**

# 1. Question 7 · part b · 5 marks

June 2024 Paper 2R

Worked-example retry · Attempt part b only.

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(iv) Describe how a student could use this equipment and a timer to compare the rate of carbon dioxide production during exercise and at rest.

**(4)**

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# 1. Question 7 · part b · 5 marks

June 2024 Paper 2R

Worked-example retry · Attempt part b only.

(b) Aerobic and anaerobic respiration both produce ATP.

Describe the role of ATP in energy transfer during exercise.

**(5)**

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**(Total for Question 7 = 14 marks)**

## 1. Question 7 · part b · 5 marks · Working and answer

Working and answer

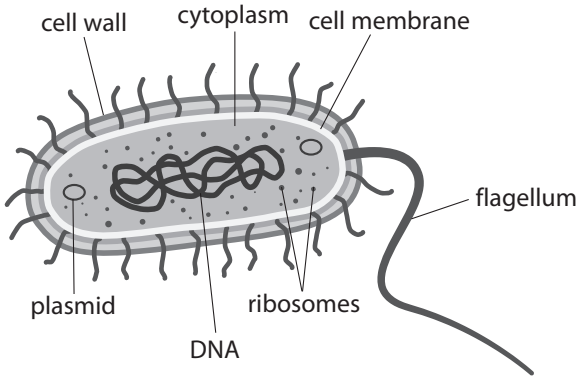
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## 2. Question 3 · part b.iii · 2 marks

January 2020 Paper 1

Further practice · Attempt part b.iii only.

**3** (a) The diagram shows a type of bacterial cell.




(i) The boxes give some structures found in the bacterial cell and some functions of cell structures.

Draw one straight line from each structure to its correct function.

**(3)**

| Structure | Function |
|---|---|
| cell membrane | protects against pathogens |
| | helps the cell to move |
| ribosomes | makes proteins |
| | holds the genetic code |
| flagellum | controls which substances enter and exit the cell |

(ii) State two structural differences between the bacterial cell and a sperm cell.

**(2)**

1 .....

.....

2 .....

.....

**8**

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## 2. Question 3 · part b.iii · 2 marks

January 2020 Paper 1

Further practice · Attempt part b.iii only.

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(b) Many bacteria absorb oxygen and nutrients from their environment. Oxygen and nutrients are used by bacteria in aerobic respiration.

(i) Which process transports oxygen into bacterial cells?

(1)

- ☒ **A** active transport
- ☒ **B** diffusion
- ☒ **C** excretion
- ☒ **D** osmosis

(ii) Complete the word equation for aerobic respiration.

(2)

oxygen + ..... → ..... + water

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(iii) Explain why respiration is important for bacteria.

(2)

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**(Total for Question 3 = 10 marks)**

## 2. Question 3 · part b.iii · 2 marks · Working and answer

Working and answer

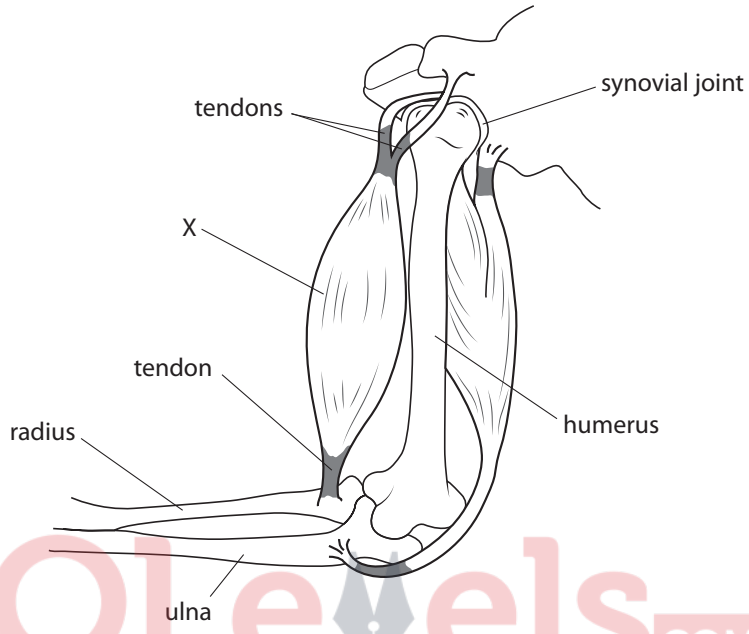
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# 3. Question 7 · part a.i · 1 mark

November 2020 Paper 1

Further practice · Attempt part a.i only.

**7** The diagram shows the bones of a human arm and the muscles that move the lower arm.




(a) (i) Muscle X needs energy to contract.

Name the process that supplies this energy.

**(1)**

.....

(ii) Muscle X contracts when the lower arm is raised further.

Describe the changes in shape of muscle X when the lower arm is raised further.

**(2)**

.....

.....

.....

.....

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P 6 2 0 5 1 A 0 2 0 2 4

## 3. Question 7 · part a.i · 1 mark · Working and answer

Working and answer

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## 4. Question 5 · part a.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part a.ii only.

**5** Some athletes train at high altitudes to improve their performance.

Performance improves if more energy is released from aerobic respiration.

(a) (i) Complete the balanced symbol equation for aerobic respiration.

**(2)**

..... + $6O_2$ → ..... + $6CO_2$

(ii) Aerobic respiration produces ATP.

Describe how energy is released from ATP during exercise.

**(2)**

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# 4. Question 5 · part a.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part a.ii only.

(b) The table shows the percentage change in mass of haemoglobin in the blood after different periods of time training at altitude.

| Period of time in hours | Percentage change in mass of haemoglobin |
|---|---|
| 0 | 0 |
| 100 | 0.7 |
| 200 | 2.4 |
| 300 | 4.1 |
| 400 | 5.9 |
| 500 | 6.5 |
| 600 | 7.2 |

(i) Plot the results on the grid.

**(4)**

(ii) Draw the curve of best fit.

**(1)**

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## 4. Question 5 · part a.ii · 2 marks · Working and answer

Working and answer

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## 5. Question 4 · part a.iv · 1 mark

June 2023 Paper 2

Further practice · Attempt part a.iv only.

**4** (a) (i) Complete the word equation for one type of respiration.

**(2)**

oxygen + ..... → ..... + .....

(ii) State the type of respiration shown by the equation.

**(1)**

.....

(iii) In which part of the cell does this type of respiration occur?

**(1)**

- ☒ **A** endoplasmic reticulum
- ☒ **B** mitochondrion
- ☒ **C** nucleus
- ☒ **D** ribosome

(iv) Name the molecule used to transfer energy in a cell.

**(1)**

.....

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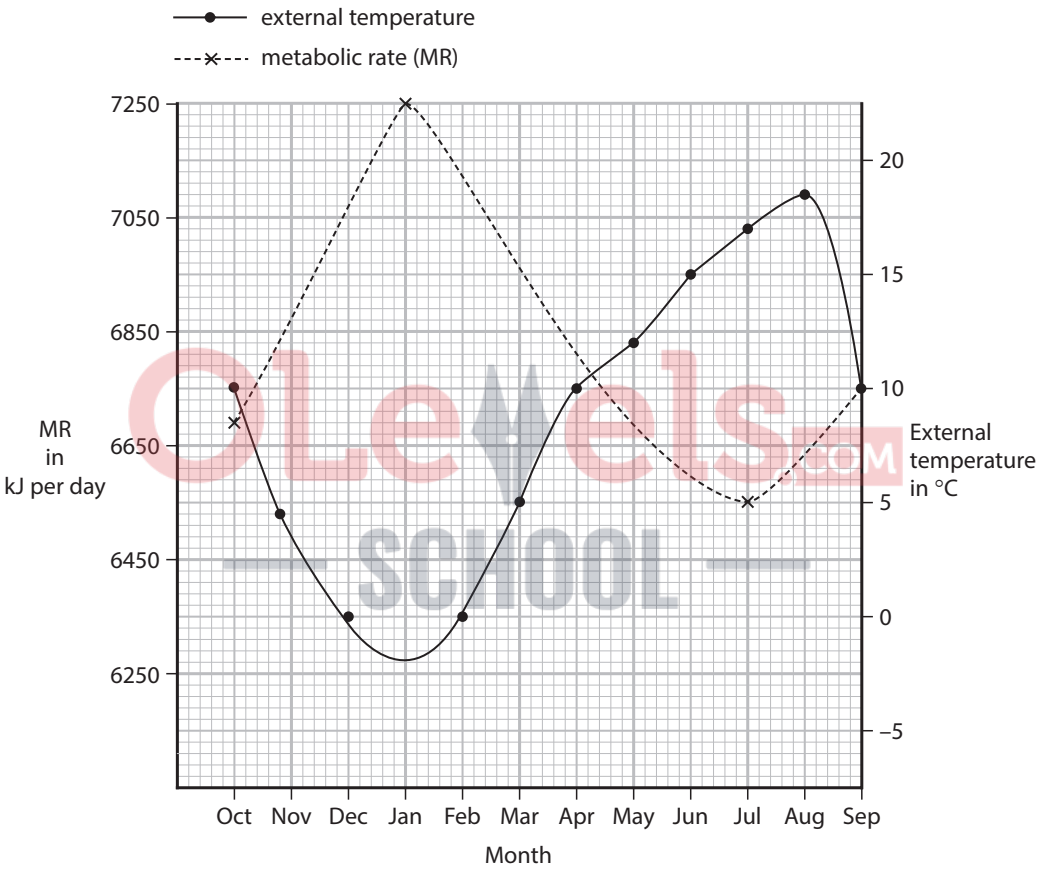
# 5. Question 4 · part a.iv · 1 mark

June 2023 Paper 2

Further practice · Attempt part a.iv only.

(b) An investigation was carried out to see how a person's metabolic rate (MR) and the external temperature change at different times in one year.

The graph shows the results of the investigation for one person.




(i) Describe how the investigation could be made more reliable.

**(2)**

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## 5. Question 4 · part a.iv · 1 mark · Working and answer

Working and answer

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# 6. Question 4 · part b.iii · 5 marks

June 2023 Paper 2

Further practice · Attempt part b.iii only.

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**4** (a) (i) Complete the word equation for one type of respiration.

**(2)**

oxygen + ..... → ..... + .....

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**(1)**

.....

(iii) In which part of the cell does this type of respiration occur?

**(1)**

- ☒ **A** endoplasmic reticulum
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- ☒ **C** nucleus
- ☒ **D** ribosome

(iv) Name the molecule used to transfer energy in a cell.

**(1)**

.....

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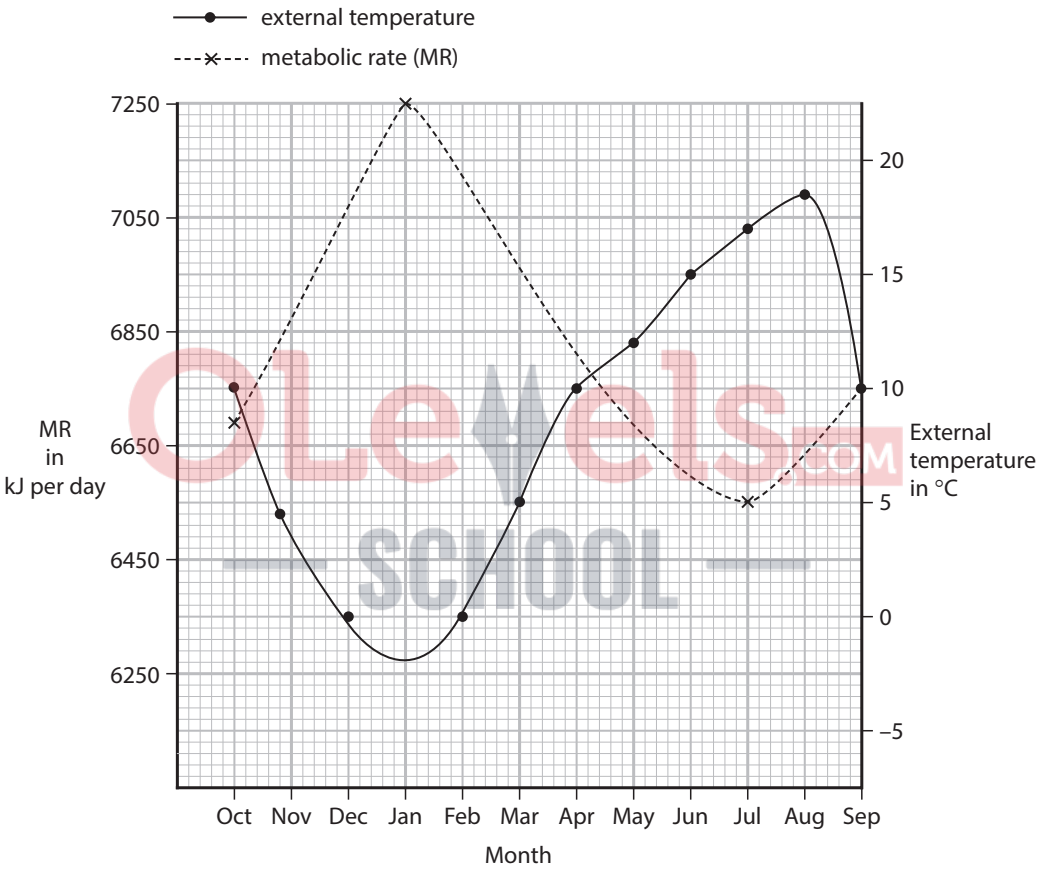
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June 2023 Paper 2

Further practice · Attempt part b.iii only.

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**(2)**

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# 6. Question 4 · part b.iii · 5 marks

June 2023 Paper 2

Further practice · Attempt part b.iii only.

(ii) Determine the difference between the maximum metabolic rate and minimum metabolic rate.

**(2)**

difference = ..... kJ per day

(iii) Explain why the metabolic rate changes throughout the year.

**(5)**

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## 6. Question 4 · part b.iii · 5 marks

June 2023 Paper 2

Further practice · Attempt part b.iii only.

(iv) Suggest how the metabolic rate could be measured.

**(1)**

.....

.....

**(Total for Question 4 = 15 marks)**

DO NOT WRITE

Working and answer

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# 7. Question 4 · part b.iv · 1 mark

June 2023 Paper 2

Further practice · Attempt part b.iv only.

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**4** (a) (i) Complete the word equation for one type of respiration.

**(2)**

oxygen + ..... → ..... + .....

(ii) State the type of respiration shown by the equation.

**(1)**

.....

(iii) In which part of the cell does this type of respiration occur?

**(1)**

- ☒ **A** endoplasmic reticulum
- ☒ **B** mitochondrion
- ☒ **C** nucleus
- ☒ **D** ribosome

(iv) Name the molecule used to transfer energy in a cell.

**(1)**

.....

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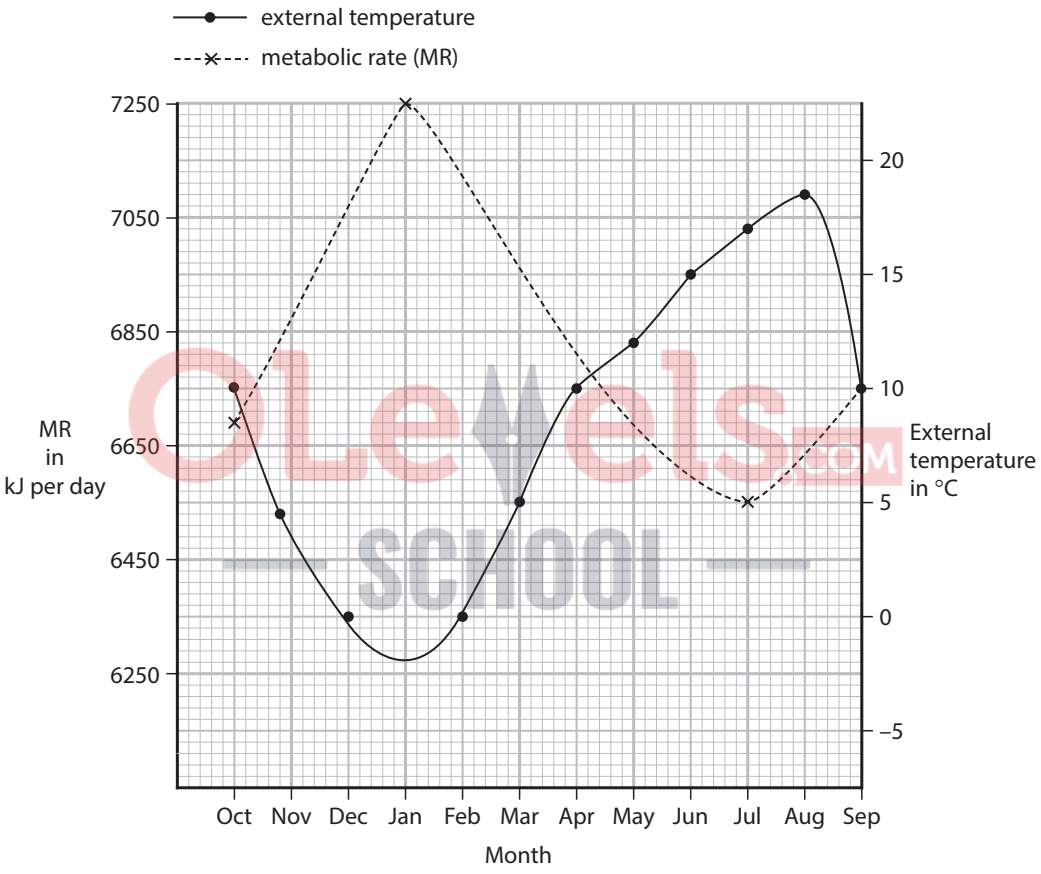
# 7. Question 4 · part b.iv · 1 mark

June 2023 Paper 2

Further practice · Attempt part b.iv only.

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(i) Describe how the investigation could be made more reliable.

**(2)**

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# 7. Question 4 · part b.iv · 1 mark

June 2023 Paper 2

Further practice · Attempt part b.iv only.

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(ii) Determine the difference between the maximum metabolic rate and minimum metabolic rate.

**(2)**

difference = ..... kJ per day

(iii) Explain why the metabolic rate changes throughout the year.

**(5)**

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# 7. Question 4 · part b.iv · 1 mark

June 2023 Paper 2

Further practice · Attempt part b.iv only.

(iv) Suggest how the metabolic rate could be measured.

**(1)**

.....

.....

**(Total for Question 4 = 15 marks)**

DO NOT WRITE

Working and answer

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Practice workbook · Set up and explain the indicator practical comparing carbon dioxide in breath

Two-tube limewater or indicator apparatus with a mouthpiece: drawing and using it, naming the indicator, and explaining results for inspired, expired, rest or exercise air.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-BREATH-CO2-INDICATOR-PRACTICAL

# 1. Question 5 · part a.ii · 3 marks

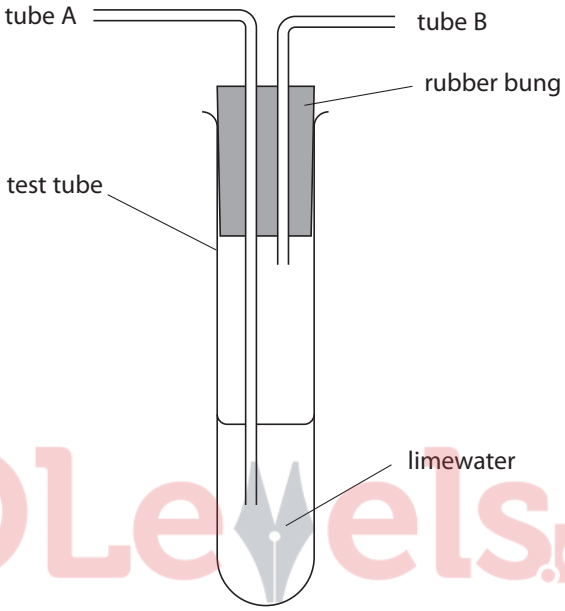
November 2021 Paper 1

Worked-example retry · Attempt part a.ii only.

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**5** The amount of carbon dioxide in inhaled air is different from the amount of carbon dioxide in exhaled air.

The diagram shows a piece of apparatus that can be used to investigate this difference.




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(a) (i) Describe how this apparatus should be used to compare the amount of carbon dioxide in inhaled and exhaled air.

**(4)**

# 1. Question 5 · part a.ii · 3 marks

November 2021 Paper 1

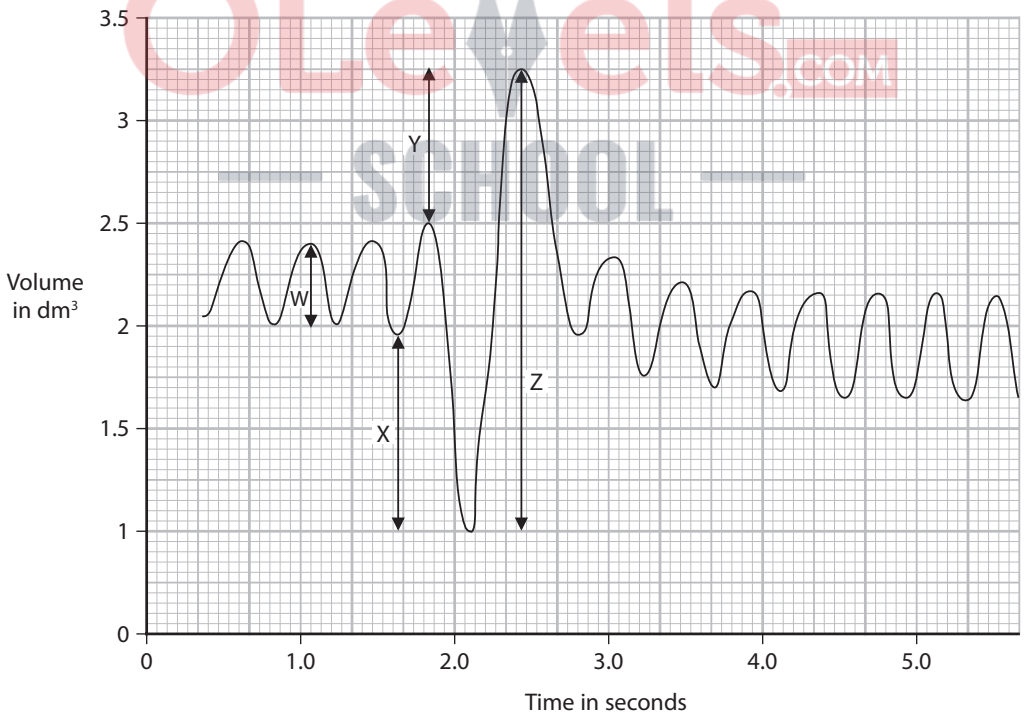
Worked-example retry · Attempt part a.ii only.

(ii) Explain the difference expected between the results.

**(3)**

(b) The movement of air in and out of the lungs can be measured using a spirometer.

The diagram shows a trace produced by a spirometer.




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## 1. Question 5 · part a.ii · 3 marks · Working and answer

Working and answer

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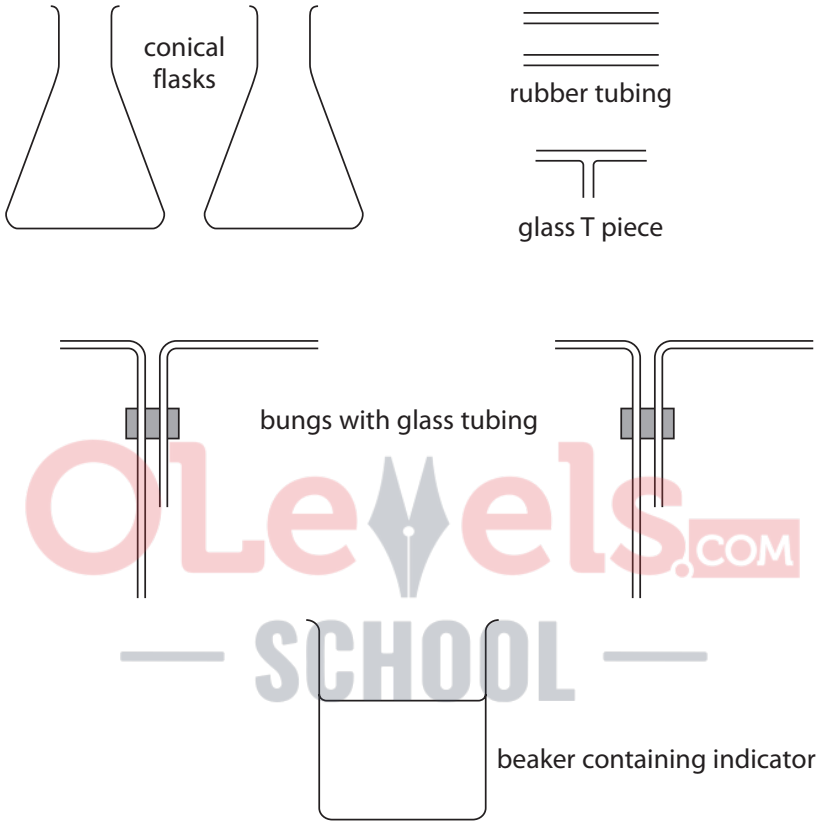
## 2. Question 1 · part a · 3 marks

June 2022 Paper 2R

Further practice · Attempt part a only.

**1** A student investigates whether there is more carbon dioxide in expired air than in inspired air.

This is the apparatus the student uses.




The apparatus is not drawn to scale.

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## 2. Question 1 · part a · 3 marks

June 2022 Paper 2R

Further practice · Attempt part a only.

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(a) Draw a diagram to show how the apparatus should be put together for the investigation.

**(3)**

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(b) Name an indicator that could be used in the investigation.

**(1)**

.....

(c) Describe how the apparatus would be used to determine whether there is more carbon dioxide in expired air than in inspired air.

**(4)**

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## 2. Question 1 · part a · 3 marks · Working and answer

Working and answer

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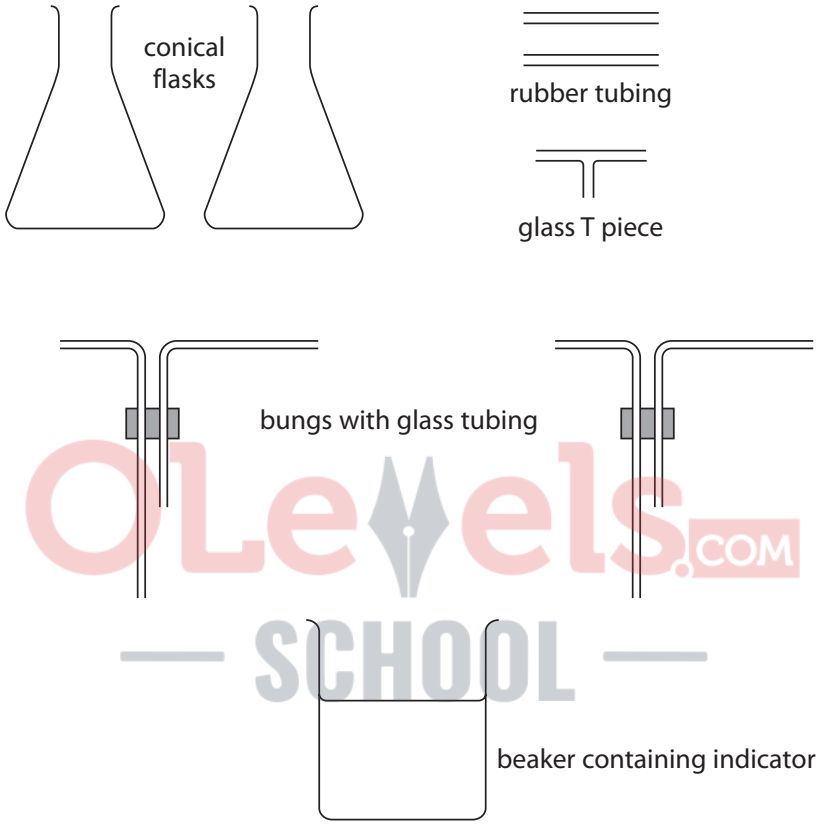
## 3. Question 1 · part b · 1 mark

June 2022 Paper 2R

Further practice · Attempt part b only.

**1** A student investigates whether there is more carbon dioxide in expired air than in inspired air.

This is the apparatus the student uses.




The apparatus is not drawn to scale.

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## 3. Question 1 · part b · 1 mark

June 2022 Paper 2R

Further practice · Attempt part b only.

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(a) Draw a diagram to show how the apparatus should be put together for the investigation.

**(3)**

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(b) Name an indicator that could be used in the investigation.

**(1)**

.....

(c) Describe how the apparatus would be used to determine whether there is more carbon dioxide in expired air than in inspired air.

**(4)**

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## 3. Question 1 · part b · 1 mark · Working and answer

Working and answer

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# 4. Question 2 · part b.i · 1 mark

June 2023 Paper 1

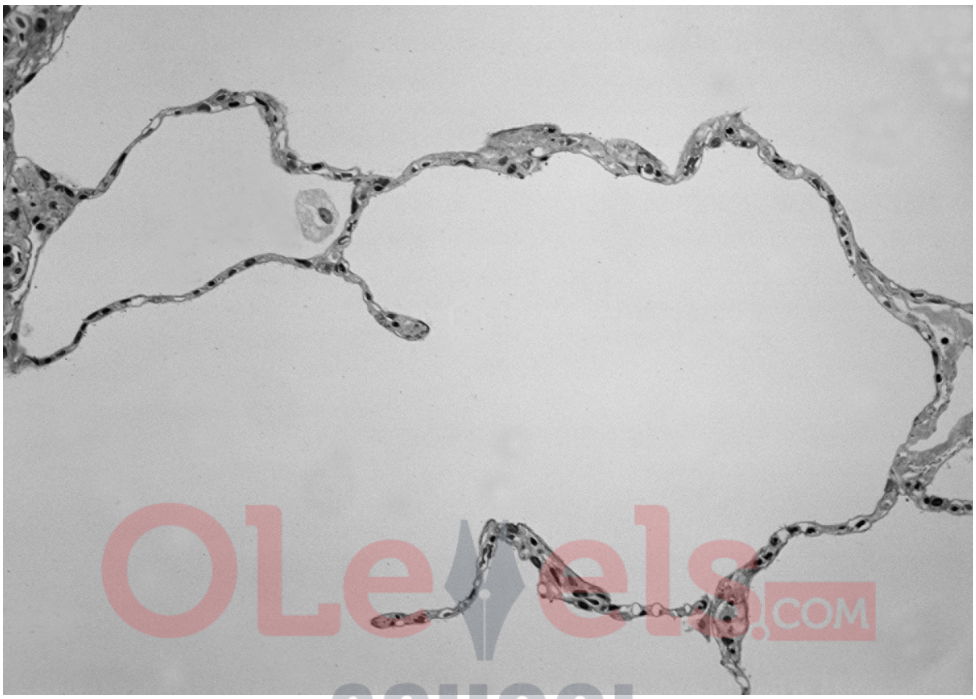
Further practice · Attempt part b.i only.

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**2** The photograph shows alveoli in the lungs. Alveoli are the site of exchange of gasses between the air and the blood.



(Source: © Ed Reschke/Getty Images)

(a) (i) State, in the correct order, the structures that air must pass through from the atmosphere to the alveoli.

**(2)**

.....

.....

(ii) State two features, shown in the diagram, that allow the efficient exchange of gases.

**(2)**

1 .....

.....

2 .....

.....

# 4. Question 2 · part b.i · 1 mark

June 2023 Paper 1

Further practice · Attempt part b.i only.

(b) (i) Which is the percentage of carbon dioxide in the atmosphere?

**(1)**

- ☒ **A** 0.01
- ☒ **B** 0.04
- ☒ **C** 0.10
- ☒ **D** 0.50

(ii) The table shows the results obtained when a person at rest breathes in air that contains different percentages of carbon dioxide.

| **Percentage (%) carbon dioxide in air** | 0.02 | 1.60 | 3.00 | 6.00 |
|---|---|---|---|---|
| **Tidal volume in $cm^3$** | 500 | 700 | 1200 | 2100 |
| **Rate of breathing in breaths per minute** | 14 | 15 | 16 | 28 |

Calculate the volume of air in $dm^3$ breathed in per minute when the person breathes air containing 3.00% carbon dioxide.

[1 $dm^3$ = 1000 $cm^3$]

**(2)**

volume of air = ..... $dm^3$

(iii) The air a person breathes in contains 20% oxygen. The body uses 20% of this oxygen.

Calculate the volume of oxygen used per minute when the person breathes air containing 1.60% carbon dioxide.

**(3)**

volume of oxygen = ..... $cm^3$

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P 7 2 5 8 5 A 0 4 2 0

## 4. Question 2 · part b.i · 1 mark · Working and answer

Working and answer

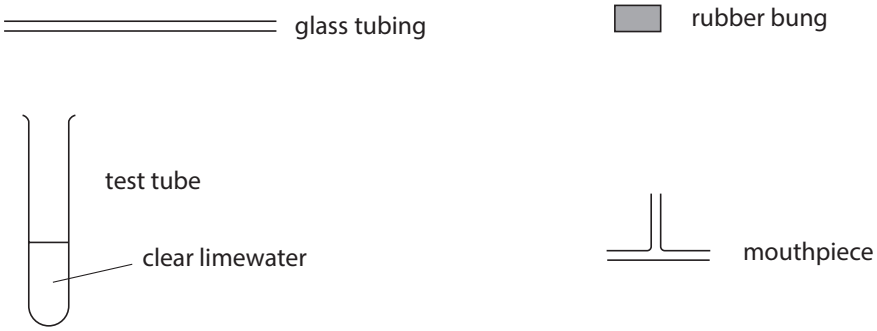
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## 5. Question 4 · part a.i · 3 marks

June 2024 Paper 1

Further practice · Attempt part a.i only.

**4** (a) The diagram shows some scientific equipment.




(i) Draw a diagram to show how this equipment is set up to investigate the difference between the carbon dioxide concentration of inspired air and of expired air.

**(3)**

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## 5. Question 4 · part a.i · 3 marks

June 2024 Paper 1

Further practice · Attempt part a.i only.

(ii) State one safety precaution in this investigation.

**(1)**

(b) A student investigates their breathing rate at rest, after gentle exercise and again after vigorous exercise.

The student repeats each exercise three times.

The student's mean results are shown in the table.

| Mean breathing rate in breaths per minute | | |
|---|---|---|
| **at rest** | **after gentle exercise** | **after vigorous exercise** |
| 14 | 19 | 50 |

(i) Give one control variable in this investigation.

**(1)**

(ii) Explain how the breathing rate after vigorous exercise benefits working muscles.

**(4)**

**(Total for Question 4 = 9 marks)**

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## 5. Question 4 · part a.i · 3 marks · Working and answer

Working and answer

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## 6. Question 7 · part a.iii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part a.iii only.

**7** (a) Aerobic and anaerobic respiration both take place during exercise.

(i) Give the word equation for anaerobic respiration.

**(1)**

.....

(ii) Complete and balance the chemical symbol equation for aerobic respiration.

**(2)**

$C_6H_{12}O_6$ + ..... ⟶ ..... + ..... $CO_2$

(iii) A student investigates the rate of carbon dioxide production during exercise and at rest.

This is the equipment they use.

- four pieces of glass tubing
- two boiling tubes
- two rubber bungs
- mouthpiece
- limewater

Draw a diagram to show how the student should set up the equipment to carry out the investigation.

**(2)**

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**20**

P 7 4 4 9 5 A 0 2 0 2 8

## 6. Question 7 · part a.iii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part a.iii only.

(iv) Describe how a student could use this equipment and a timer to compare the rate of carbon dioxide production during exercise and at rest.

**(4)**

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## 6. Question 7 · part a.iii · 2 marks · Working and answer

Working and answer

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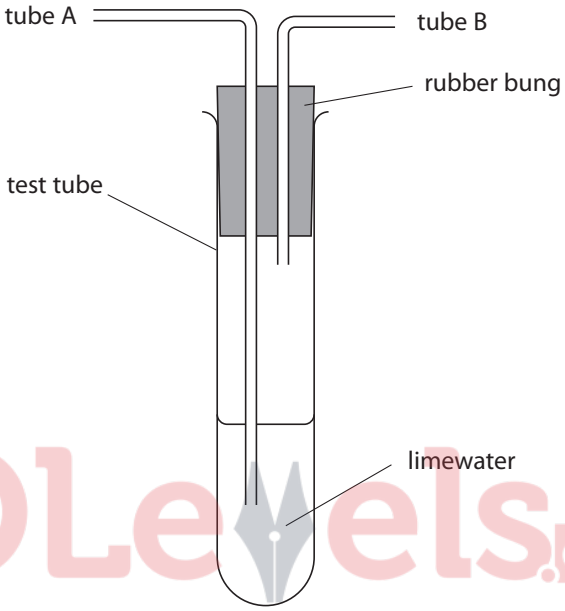
## 7. Question 5 · part a.i · 4 marks

November 2021 Paper 1

Further practice · Attempt part a.i only.

**5** The amount of carbon dioxide in inhaled air is different from the amount of carbon dioxide in exhaled air.

The diagram shows a piece of apparatus that can be used to investigate this difference.




(a) (i) Describe how this apparatus should be used to compare the amount of carbon dioxide in inhaled and exhaled air.

**(4)**

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## 7. Question 5 · part a.i · 4 marks

November 2021 Paper 1

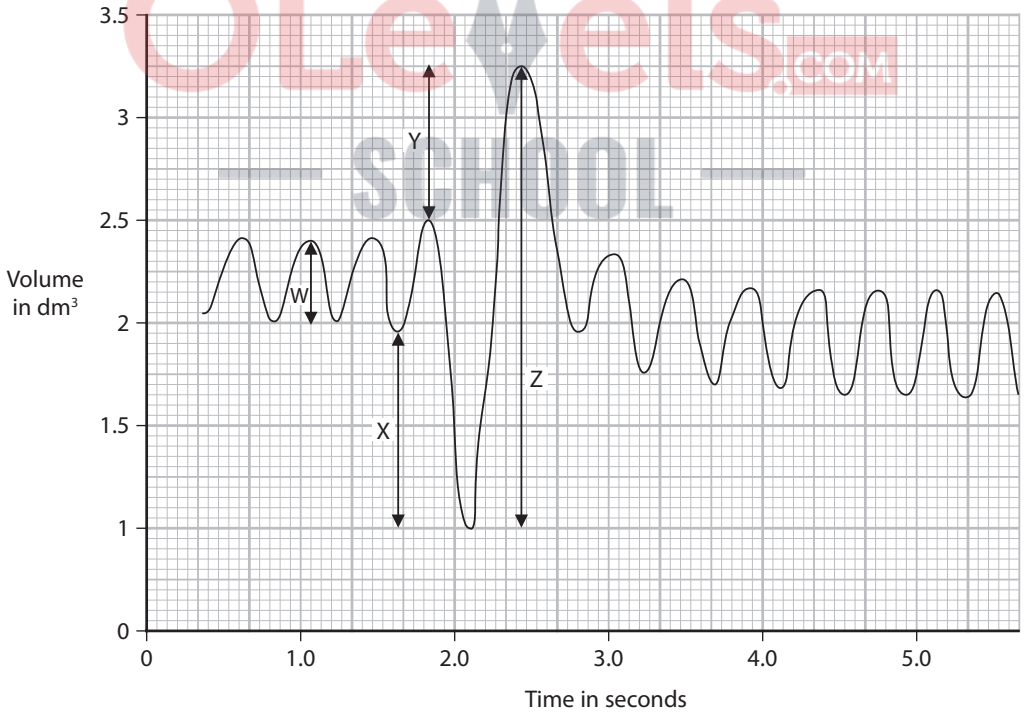
Further practice · Attempt part a.i only.

(ii) Explain the difference expected between the results.

**(3)**

(b) The movement of air in and out of the lungs can be measured using a spirometer.

The diagram shows a trace produced by a spirometer.




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## 7. Question 5 · part a.i · 4 marks · Working and answer

Working and answer

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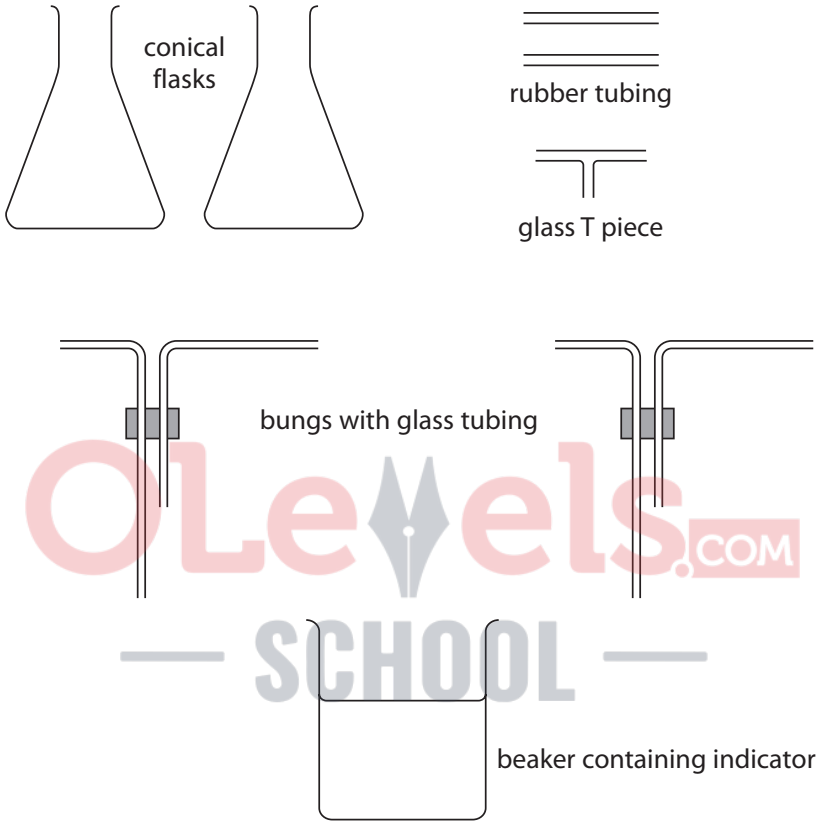
# 8. Question 1 · part c · 4 marks

June 2022 Paper 2R

Further practice · Attempt part c only.

**1** A student investigates whether there is more carbon dioxide in expired air than in inspired air.

This is the apparatus the student uses.




The apparatus is not drawn to scale.

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## 8. Question 1 · part c · 4 marks

June 2022 Paper 2R

Further practice · Attempt part c only.

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(a) Draw a diagram to show how the apparatus should be put together for the investigation.

**(3)**

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(b) Name an indicator that could be used in the investigation.

**(1)**

(c) Describe how the apparatus would be used to determine whether there is more carbon dioxide in expired air than in inspired air.

**(4)**

## 8. Question 1 · part c · 4 marks

June 2022 Paper 2R

Further practice · Attempt part c only.

(d) State one safety precaution.

**(1)**

.....

.....

(e) State one variable that should be controlled.

**(1)**

.....

.....

**(Total for Question 1 = 10 marks)**

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Working and answer

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# 9. Question 7 · part a.iv · 4 marks

June 2024 Paper 2R

Further practice · Attempt part a.iv only.

**7** (a) Aerobic and anaerobic respiration both take place during exercise.

(i) Give the word equation for anaerobic respiration.

**(1)**

.....

(ii) Complete and balance the chemical symbol equation for aerobic respiration.

**(2)**

$C_6H_{12}O_6$ + ..... ⟶ ..... + ..... $CO_2$

(iii) A student investigates the rate of carbon dioxide production during exercise and at rest.

This is the equipment they use.

- four pieces of glass tubing
- two boiling tubes
- two rubber bungs
- mouthpiece
- limewater

Draw a diagram to show how the student should set up the equipment to carry out the investigation.

**(2)**

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**20**

## 9. Question 7 · part a.iv · 4 marks

June 2024 Paper 2R

Further practice · Attempt part a.iv only.

(iv) Describe how a student could use this equipment and a timer to compare the rate of carbon dioxide production during exercise and at rest.

**(4)**

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## 9. Question 7 · part a.iv · 4 marks

June 2024 Paper 2R

Further practice · Attempt part a.iv only.

(b) Aerobic and anaerobic respiration both produce ATP.

Describe the role of ATP in energy transfer during exercise.

**(5)**

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**(Total for Question 7 = 14 marks)**

## 9. Question 7 · part a.iv · 4 marks · Working and answer

Working and answer

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